

Question ID: 3ea87153

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question

The function g is defined by $g(x) = x^2 + 9$. For which value of x is $g(x) = 25$?

Answer

- A. 4
- B. 5
- C. 9
- D. 13

Correct Answer: A

Rationale

Choice A is correct. It's given that $g(x) = x^2 + 9$. Substituting **25** for $g(x)$ in this equation yields $25 = x^2 + 9$. Subtracting **9** from both sides of this equation yields $16 = x^2$. Taking the square root of each side of this equation yields $x = \pm 4$. It follows that $g(x) = 25$ when the value of x is **4** or **−4**. Only **4** is listed among the choices.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.